Sang-Kyun Ko (고상균, 高像均)

Uiwang-si, Gyeonggi-do, Republic of Korea | highsg19101@gmail.com | +82 10-5153-4019

Google Scholar | LinkedIn | Personal Website | Academic Statement

RESEARCH PROFILE

Machine learning researcher specializing in medical AI, computer vision, and flow-based generative modeling for coronary angiography and endoscopic imaging. Experienced in academic-hospital collaboration, applied research aimed at developing deployable products, and model development across image restoration, video frame interpolation, tracking, survival analysis, and cardiovascular risk prediction.

CORE COMPETENCIES

- **Medical AI and Computer Vision:** coronary angiography analysis, endoscopic image analysis, image restoration, object tracking, video frame interpolation, 3D reconstruction
- **Generative Modeling:** diffusion models, rectified flow, flow matching, neural fields, robustness under distribution shift
- **Predictive Modeling:** cardiovascular risk prediction, survival analysis with censored data, concordance index optimization

RESEARCH INTERESTS

- Flow-based generative modeling for medical imaging, video generation, and representation learning under distribution shift
- Addressing theoretical and statistical issues in machine learning for large-scale data analysis

PROFESSIONAL EXPERIENCE

Machine Learning Researcher Oct 2023 – Present

Seoul National University Bundang Hospital (SNUBH), Coronary Artery AI Team, Cardiovascular Center, Internal Medicine

Research contract with professor

- Lead research on CAG image restoration from fluoroscopy to cine-quality sequences using flow- and diffusion-based models.
- Develop automated coronary artery tracking methods in CAG sequences for downstream analysis and clinical visualization support.
- Contributed to latent flow matching-based video frame interpolation for coronary angiography; related manuscript published in *npj Digital Medicine*.
- Led an atherosclerotic cardiovascular disease (ASCVD) risk prediction project using Kolmogorov-Arnold Networks (KANs) and health screening data.
- Investigate 3D reconstruction of coronary arteries from CAG sequences to improve anatomical interpretation and visualization.

Machine Learning Research Engineer Apr 2022 – Jul 2023
AINEX, Gastroscopic AI Part, AI Research and Development Team
Full-time employee

- Researched diffusion model-based endoscopic image generation for data augmentation and medical image analysis workflows.
- Developed gastroscopic image analysis models for diagnosis-support tasks, including real-time image classification.
- Worked on monocular object-size estimation for detected objects in endoscopic images.
- Collaborated with the AI R&D team on model experimentation, error analysis, and deployment-oriented validation.

Assistant Researcher Apr 2019 – Jul 2019
Industry-University Cooperation Foundation, Sun Moon University, System Software Research Lab.
Contract employee

- Developed CNN-based Chinese cursive character recognition methods for a university-industry research project.

PUBLICATIONS

1. **Ko, S.-K.**, Kim, S.-H., Kim, S.-H., Kim, H., Yoon, C.-H., Youn, T.-J., & Chae, I.-H. (2026). Kolmogorov-Arnold Network for Atherosclerotic Cardiovascular Disease Risk Prediction. *medRxiv*. [Under review]
2. Kwon, H., Park, S., Kim, D. H., **Ko, S.-K.**, Kim, S. H., Moon, I. T., et al. (2026). Video frame interpolation for coronary angiography using latent flow matching. *npj Digital Medicine*.
3. Moon, I. T.*, **Ko, S.-K.***, Kang, S. H., et al. (2024). Augmented Reality in Cardiology: Enhancing Visualization and Precision. *Current Cardiovascular Risk Reports*. *Equal contribution.
4. Jang, C., **Ko, S.**, Choi, J., Lim, J., Noh, Y. K., & Kim, T. J. (2022). Learning to increase matching efficiency in identifying additional b-jets in the $t\bar{t}b\bar{b}$ process. *The European Physical Journal Plus, 137*(7), 1–12.

CONFERENCES

1. **Ko, S.-K.**, Kang, S. H., Kim, S. H., Kim, H., Youn, T. J., Yoon, C. H., & Chae, I. H. (2026). Kolmogorov-Arnold Network for Atherosclerotic Cardiovascular Disease Risk Prediction. American College of Cardiology Scientific Session 2026 (ACC.26), Abstract Presentation. Published in *Journal of the American College of Cardiology, 87*(13 Suppl), A43. [Poster] [Project page]
2. **Ko, S.-K.** & Noh, Y.-K. (2023). Data Reconstruction Method for Learning to Censored Survival Data. *Korean Artificial Intelligence Association Summer Conference 2023*. Poster presentation. [Program book] [Paper] [Poster]
3. Oh, M. J., Park, J., Jeon, J., **Ko, S.-K.**, Jo, S., Kang, S., Kim, S. H., Park, S. H., Chang, Y. H., Shin, C. M., Kang, S. J., Kim, S. G., & Cho, S. J. (2023). Application of artificial intelligence in the detection of Borrmann type 4 advanced gastric cancer in upper endoscopy. *United European Gastroenterology Journal, 11*(Supplement 8), pp0253, p666. Poster presentation. [Abstract]

4. **Ko, S.**, Kim, B., & Kim, J. D. (2018). Deep Learning-based Algorithm for Object Identification in Multimedia. *The 13th KIPS International Conference on Ubiquitous Information Technologies and Applications* (*CUTE 2018*). Oral presentation. [Program book]

5. Park, K., Hong, B., **Ko, S.**, & Kim, B. (2018). Design and Implementation of Real-Time Web Authoring Tool Based on Drag-and-Drop Method. *The 13th KIPS International Conference on Ubiquitous Information Technologies and Applications* (*CUTE 2018*). Oral presentation. [Program book]

6. Seo, H., Hong, B., **Ko, S.-K.**, Park, K., Kim, B., & Jung, H. (2017). A survey on the performance of deep learning based on multiple GPUs. *Korea Software Congress 2017*. Poster presentation. [Program book] [Paper]

EDUCATION

Hanyang University (HYU), Seoul, Korea Sep 2019 – Feb 2022; Sep 2023 – Feb 2024
Advanced Application for Intelligence Systems Lab. (aAIs Lab), Department of Computer Science [Lab link]

- M.S. in Computer Science; advisor: Prof. Yung-Kyun Noh
- Thesis: "Learning to Directly Maximize Evaluation Metrics: Focus on Matching Efficiency and Concordance Index" [Thesis]
- Teaching Assistant: Artificial Intelligence (CSE4007), Automata Theory (ITE3061), Discrete Mathematics (MAT2020), Creative Software Programming: C++ (ITE1015)
- Led research on utilizing censored data as training data in survival analysis.
- Collaborated with particle physics researchers on identifying additional b-jets in the $t\bar{t}b\bar{b}$ process.

Sun Moon University, Asan, Korea Mar 2013 – Feb 2019
System Software Research Lab. (SSR Lab), Department of Computer Science
Later renamed Intelligent System and Software Research Lab., Chungbuk National University [Lab link]

- B.S. in Computer Science; advisor: Prof. Bongjae Kim
- Graduation project: "Deep Learning-based System for Object Retrieval in Multimedia"
- Student Excellence Scholarship, 2017

SHORT TERM EDUCATION

- Pattern Recognition and Machine Learning Winter School 2024, AI Society, Feb 2024 (12 hours, online). [Link]
- Pattern Recognition and Machine Learning Winter School 2021, AI Society, Feb 2021 (8 hours, online). [Link]
- The 11th KIAS CAC Online Winter School: Parallel Computing and Artificial Intelligence, Dec 2020 (20 hours, online). [Link]

AWARDS AND HONORS

- Nov 2018, **Excellence Prize**, 4th Industrial Revolution Talent Development Joint Festival, paper/coding division, Korea Society of Information Communication Security Ethics

- Nov 2018, **Grand Prize**, Comprehensive Competition for Computer Science and Electrical Engineering, Department of Computer Science, Sun Moon University
- Jun 2018, **Silver Prize**, Comprehensive Competition for Computer Science, Department of Computer Science, Sun Moon University
- Nov 2017, **Bronze Prize**, Global Capstone Design Workshop, Idea Factory, Sun Moon University
- Apr 2017, **Excellence Prize**, Idea Contest held by the President, Sun Moon University

CERTIFICATES

- Engineer Information Processing (정보처리기사), Human Resources Development Service of Korea, May 2019

MILITARY SERVICE

- Mar 2014 – Dec 2015, Republic of Korea Army, Infantry, Yanggu-gun, Gangwon-do; completed mandatory military service as a platoon signalman.

REFERENCES

- **Yung-Kyun Noh** (nohyung@hanyang.ac.kr)
  Professor, Computer Science, Hanyang University
- **Bongjae Kim** (bjkim@chungbuk.ac.kr)
  Professor, Computer Science, Chungbuk National University